A Project Synopsis Of

AI EXAM NOTES GENERATOR

Submitted for the partial fulfillment of the requirement for the degree of

Bachelor of Computer Applications (BCA)
Academic Session- 2023-2026
Department of Computer Applications

Submitted By:

Tanushree Dutta (Reg No: NSU2303265)
Payal Mahakur (Reg No: NSU2303163)

Under the Guidance of

Ritesh Kumar Jha
Assistant Professor (CS & IT)
Netaji Subhas University Jamshedpur

ACKNOWLEDGEMENT

We would like to express our sincere gratitude and heartfelt thanks to all those who have directly or indirectly contributed to the successful completion of our project titled "**AI Exam Notes Generator**", submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Computer Applications (BCA)** during the academic session **2023–2026**. This project would not have been possible without the continuous support, guidance and encouragement of many respected individuals.

First and foremost, we express our deep sense of gratitude to our respected **Project Guide, Mr. Ritesh Kumar Jha**, for his valuable guidance, constant encouragement, constructive suggestions and continuous support throughout the development of this project. His knowledge, patience and motivation helped us to understand the project requirements clearly and complete the work successfully within the stipulated time.

We would also like to extend our sincere thanks to our respected **Chancellor Sir, Registrar Sir, Dean Sir and Head of the Department (HOD)** for their leadership, administrative support and encouragement. Their vision, academic guidance and commitment to providing quality education created a positive learning environment that enabled us to complete this project successfully. We are also thankful to all faculty members and staff of the Department of Computer Applications for providing the necessary facilities, cooperation and valuable suggestions during the preparation of this project.

We extend our sincere thanks to Ardent Computech Pvt. Ltd. and TechCoder Software (OPC) Pvt. Ltd. for providing us with industrial internship opportunities, practical industry exposure and technical mentorship that greatly enriched our understanding of real-world software development.

Finally, we are deeply grateful to our parents, family members, friends and the management and staff of our internship organizations for their continuous encouragement, moral support and cooperation throughout our academic journey. Their support and motivation played an important role in the successful completion of this project. We sincerely acknowledge the contribution of everyone who helped us in any way during the development of this project.

DECLARATION

We hereby declare that the project entitled "**AI Exam Notes Generator**" submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Computer Applications (BCA) to NSU,** Jamshedpur, is a record of original work carried out by us under the guidance and supervision of **Mr. Ritesh Kumar Jha, Project Guide, Department of Computer Applications.**

We further declare that this project work has been completed by us independently and has not been submitted, either in part or in full, to any other university or institution for the award of any degree, diploma, or certificate.

We also confirm that all the information, data and materials used in this project have been properly acknowledged and referenced wherever required. We have complied with all academic rules, regulations and ethical standards prescribed by the institution. Any assistance received during the development of this project has been duly acknowledged in the acknowledgement section of this report.

We understand that any act of plagiarism or academic dishonesty, if found at any stage, may lead to cancellation of our project and disciplinary action as per the rules of the university.

This project represents our genuine effort and learning during the course of study in the **Bachelor of Computer Applications (BCA)** program for the **academic session 2023–2026.**

Student Signatures:

S.No.	Name	Registration No.	Section	Signature
1	Tanushree Dutta	NSU2303265	B	
2	Payal Mahakur	NSU2303163	A	

Project Guide:
Name: Ritesh Kumar Jha
Signature: _____
Date: _____ **Place:** Jamshedpur

INDEX

S. No.	Contents	Page No.
1	Acknowledgement	2
2	Declaration	3
3	Title of the Project	5
4	Introduction	6
5	Objectives of the Project	7
6	Problem Statement	8
7	Scope of the Project	9
8	Methodology / Technology Used	10-11
9	Hardware and Software Requirements	12-13
10	Expected Outcome	14
11	Conclusion	15
12	Data Flow Diagram (DFD) – Level 0 (Context Diagram)	16
13	Data Flow Diagram (DFD) – Level 1	17-18
14	ER Diagram (Entity Relationship Diagram)	19
15	Entities, Attributes and Relationships	20-21
16	Database Structure	22-23
17	References / Bibliography	24
18	Internship Details	25
19	Internship Certificate	26-27

TITLE OF THE PROJECT

AI EXAM NOTES GENERATOR

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Computer Applications (BCA)
Academic Session: 2023 – 2026

Submitted By

S. No.	Name	Registration No.	Section	Contact
1	Tanushree Dutta	NSU2303265	B	8789782465
2	Payal Mahakur	NSU2303163	A	9608884199

Course Details

Course Name: Bachelor of Computer Applications (BCA)

Semester: 6th Semester

Department: Computer Applications

Under the Guidance of

Guide Name: Ritesh Kumar Jha

Designation: Assistant Professor (CS & IT)

Submitted To

Department of Computer Applications
NSU, Jamshedpur
Date: _____ **Place:** Jamshedpur

INTRODUCTION

The **AI Exam Notes Generator** is an innovative web-based Software as a Service (SaaS) application that leverages Artificial Intelligence to automatically generate structured, exam-oriented study notes for students. Built using the **MERN stack (MongoDB, Express.js, React.js, Node.js)** the platform integrates the Google Gemini AI API to produce high-quality notes, diagrams, charts and revision points based on any topic provided by the user.

In today's fast-paced academic environment, students often struggle with the time-consuming process of manually preparing notes from textbooks and lecture materials. Traditional note-making methods require significant time and effort, leaving students with insufficient time for revision and practice. Furthermore, identifying the most important and exam-relevant topics from lengthy study materials has always been a major challenge for learners preparing for competitive and university examinations.

The AI Exam Notes Generator directly addresses these problems by providing an intelligent, AI-powered assistant that generates topic-specific, exam-oriented notes within minutes. Students simply enter a topic, their class level and their examination type and the system automatically produces comprehensive notes tailored to their needs. The generated content includes detailed topic notes, priority-based revision points, block diagrams, bar charts, pie charts, line charts, short answer questions, long answer questions and diagram- based questions all structured for maximum exam readiness.

The platform follows a credit-based monetization model using the Stripe Payment Gateway, making it a fully functional, commercial-grade SaaS product. New users receive 100 free credits upon registration and can purchase additional credits as needed. The system uses Firebase for secure Google OAuth 2.0 authentication, MongoDB for data storage and is deployed on cloud infrastructure for global accessibility. A PDF download feature allows students to save and study their generated notes offline.

This project demonstrates the practical integration of modern web technologies, artificial intelligence and cloud-based services to create a meaningful and scalable solution for the education sector. It not only provides a valuable learning tool for students across India but also represents a production-ready SaaS application showcasing industry-standard software engineering practices.

OBJECTIVES OF THE PROJECT

The main objective of the **AI Exam Notes Generator** project is to design and develop an AI-powered web application that simplifies exam preparation for students by automatically generating structured, topic-specific notes. The system aims to provide a convenient, reliable and intelligent platform for students to generate high-quality study material in minimal time.

The specific objectives of the project are as follows:

- To generate comprehensive, AI-powered exam notes on any topic or subject instantly, tailored to the student's class level and examination type.
- To reduce the time students spend on manual note preparation by automating the content generation process using Google Gemini AI.
- To provide quick revision material including prioritized revision points, short notes, diagrams and charts to support effective last-minute exam preparation.
- To improve student learning efficiency through smart, priority-based content delivery with star-rated topic importance levels.
- To generate interactive visual content including block diagrams, bar charts, pie charts and line charts for better understanding of complex topics.
- To implement secure user authentication using Firebase Google OAuth 2.0 to ensure data privacy and user security.
- To integrate a real-world payment gateway (Stripe) for a credit-based business model, enabling the platform to function as a scalable SaaS product.
- To allow students to download their generated notes as formatted PDF documents for offline use.
- To maintain a notes history system so users can revisit and re-download all previously generated notes at any time.
- To create a scalable, commercially viable SaaS educational application that can serve students across multiple examination boards including CBSE, AKTU and others.

PROBLEM STATEMENT

In the current educational environment, students preparing for examinations face several significant challenges that affect their academic performance and overall learning experience. One of the most common and critical problems is the excessive amount of time required to manually prepare study notes from lengthy textbooks and lecture materials. Students often spend hours compiling information, summarizing chapters and organizing notes, leaving very little time for actual revision and practice. This problem is particularly severe during examination periods when time is of the essence.

Another major challenge faced by students is the difficulty in identifying the most important and exam-relevant topics from large volumes of study material. Without proper guidance, students often spend equal time on all topics, including those with low examination weightage, resulting in inefficient use of their study time. Additionally, the lack of structured revision material such as quick revision points, expected questions and visual aids makes last-minute exam preparation highly stressful and ineffective.

Existing AI tools and general-purpose chatbots, while helpful for providing information, do not generate structured, exam-formatted notes that include visual elements such as diagrams and charts. They do not tailor their responses to specific class levels, examination types, or prioritize topics based on examination patterns. Students using such tools receive generic information that requires significant additional processing before it can be used as effective study material.

Furthermore, there is no readily available, accessible and affordable platform specifically designed to generate prioritized exam notes with visual diagrams, charts and exam-style questions customized for specific examination boards such as CBSE or AKTU. The absence of such a tool creates a gap in the educational technology market that this project aims to fill.

The **AI Exam Notes Generator** is designed to solve these problems by providing a fast, intelligent and structured note-generation system that produces comprehensive exam-ready content including notes, revision points, block diagrams, charts and question sets all automatically tailored to the user's specified topic, class level and examination type.

SCOPE OF THE PROJECT

The scope of the **AI Exam Notes Generator** project defines the boundaries, features and functionalities of the system and explains how it can be used in real-world academic environments. The primary scope is to develop a fully functional, cloud-deployed web application that allows students to generate AI-powered, exam-oriented study notes for any topic, class level and examination type quickly and efficiently.

The project covers the following core functionalities and areas:

- Web-based SaaS application accessible from any device via a standard web browser without requiring any software installation.
- Multi-topic support: any subject or topic from any class level (Class 10, Class 12, BCA, B.Tech, etc.) can be used to generate notes.
- Examination type customization: supports CBSE, AKTU and other examination boards, with notes tailored accordingly.
- Comprehensive content generation including: detailed topic notes, priority-based revision points (1-star, 2-star, 3-star), block diagrams, bar charts, pie charts, line charts, short answer questions, long answer questions and diagram-based questions.
- Notes history panel: all previously generated notes are saved in MongoDB and accessible for re-viewing and re-downloading at any time.
- PDF download: users can download all generated notes as a fully formatted PDF document for offline study.
- Credit-based system: users receive 100 free credits on registration; additional credits can be purchased via the Stripe payment gateway.
- Secure Google OAuth 2.0 authentication via Firebase for safe and easy user login.
- Cloud deployment on Render for global accessibility and high availability.

The scope of this project is focused on web-based note generation using AI. In the future, the system can be expanded to include features such as mobile application support, collaborative note sharing, AI-generated mock tests, performance analytics and support for regional language content. These enhancements will further increase the scalability and usability of the platform for a wider student audience across India and beyond.

METHODOLOGY / TECHNOLOGY USED

The development of the **AI Exam Notes Generator** follows a systematic and structured methodology based on the **Software Development Life Cycle (SDLC)** approach. This includes the phases of planning, analysis, design, development, testing and deployment. The project is built on the **MERN stack (MongoDB, Express.js, React.js, Node.js)** with integration of the Google Gemini AI API, Firebase authentication, Stripe payment processing and cloud deployment.

Front-End Technologies

React.js: Used to build the single-page application UI with component-based architecture, enabling dynamic and responsive user interactions.

Tailwind CSS: Used for rapid, responsive styling of the user interface with utility-first CSS classes.

JavaScript (ES6+): Used for client-side logic, event handling and API communication.

Recharts: A React-based charting library used to render bar charts, pie charts and line charts from AI-generated data.

Back-End Technologies

Node.js: Server-side JavaScript runtime environment used to build the backend application.

Express.js: A lightweight web framework used for creating RESTful API routes, middleware handling and server-side logic.

Database

MongoDB: A NoSQL document database used to store user data, generated notes and transaction records. MongoDB Atlas is used for cloud-based database hosting.

AI Integration

Google Gemini API: The AI engine used to generate structured exam notes, revision points, block diagrams, charts and exam questions based on the user's input topic, class level and exam type.

Authentication

Firebase (Google OAuth 2.0): Used for secure user authentication via Google login, enabling easy and safe access to the platform.

Payment Gateway

Stripe: Integrated for processing credit purchases. Stripe webhooks are used to confirm payment completion and automatically update user credit balances in the database.

Development Tools

- **VS Code:** Code editor used for writing and managing all project files.
- **Git & GitHub:** Used for version control and collaborative code management.
- **Postman:** Used for testing and debugging RESTful API endpoints.
- **MongoDB Compass:** Used for database inspection and query testing.
- **Render:** Cloud platform used for deploying both the frontend and backend of the application.

HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements

The hardware requirements for developing and running the **AI Exam Notes Generator** are as follows:

- **Processor:** Intel Core i3 or higher a suitable processor is required to execute program instructions efficiently and handle multiple tasks during system development and operation.
- **Random Access Memory (RAM):** Minimum 4 GB (Recommended: 8 GB) Adequate RAM ensures smooth performance, especially when running development tools, local servers and the application simultaneously.
- **Hard Disk / Storage:** Minimum 20 GB free space Sufficient storage is necessary for software, database files, project documents and system backups.
- **Monitor:** Any standard LCD / LED Monitor Required to view the user interface, program code and application output.
- **Keyboard and Mouse:** Standard input devices required for development tasks and system interaction.
- **Network Connection:** Stable Internet Connection Required for AI API calls to Google Gemini, Firebase authentication, Stripe payment processing and cloud deployment.

Software Requirements

The software requirements for developing and operating the **AI Exam Notes Generator** include the following:

- **Operating System:** Windows 10/11, macOS, or Linux. The operating system provides the platform for running development tools and the application.
- Node.js (v18 or higher) and npm required to run the backend server and manage project dependencies.
- React.js Frontend JavaScript library used for building the user interface.
- MongoDB (local or MongoDB Atlas cloud cluster) Database system for storing user data, notes and transactions.

- Google Gemini API Key required for AI-powered note generation functionality.
- Firebase Project Required for Google OAuth 2.0 authentication setup.
- Stripe Account required for payment gateway integration and webhook configuration.
- VS Code Code editor for writing and managing project files.
- Git and GitHub used for version control and source code management.
- Modern Web Browser (Google Chrome / Firefox / Edge) required to access and test the web application.
- Postman used for API testing and debugging during development.

EXPECTED OUTCOME

The **AI Exam Notes Generator** is expected to provide an efficient, reliable and intelligent platform for generating exam-ready study notes and improving academic preparation for students. The successful implementation of this system will simplify the note generation process, reduce manual effort and significantly improve the quality and speed of exam preparation.

One of the primary expected outcomes of this project is the ability to generate comprehensive, AI-powered exam notes on any topic within 2 to 5 minutes. These notes will include detailed content sections, priority-based revision points, block diagrams, bar charts, pie charts, line charts, short answer questions, long answer questions and diagram-based questions all tailored to the user's specified class level and examination type.

Another important expected outcome is the effective management of user data, generated notes and transaction records through a structured MongoDB database. The system will store all generated notes securely, allowing users to access their complete notes history, re- view previously generated content and re-download notes as formatted PDF documents at any time.

The credit-based monetization system integrated with Stripe is expected to function as a production-ready SaaS business model, enabling users to purchase credits securely and have their balances updated automatically through Stripe webhook confirmation. This makes the platform commercially viable and scalable.

The quick 5-minute revision mode is expected to provide students with a condensed set of the most important exam points for any topic, making last-minute revision fast, focused and effective. The PDF download feature will further allow students to study generated notes offline on any device.

Overall, the **AI Exam Notes Generator** is expected to serve as a practical, scalable and impactful educational tool that helps students across multiple examination boards and class levels prepare more effectively for their exams, while also demonstrating the real-world application of MERN stack development, AI integration and SaaS product design.

CONCLUSION

The **AI Exam Notes Generator** project has been successfully designed and developed to provide an efficient and intelligent solution for automated exam note generation using Artificial Intelligence. The system replaces the time-consuming process of manual note preparation with an AI-powered platform that generates comprehensive, structured and exam-ready study material within minutes. It provides a simple and intuitive user interface that allows students to generate notes, diagrams, charts and exam questions for any topic, class level and examination type.

This project demonstrates the practical integration of modern web development technologies and artificial intelligence in solving a real-world academic problem. The MERN stack provides a robust and scalable full-stack framework, while the Google Gemini AI API enables intelligent and context-aware content generation. The integration of Firebase for authentication, Stripe for payment processing, Recharts for visual chart rendering and MongoDB for data storage creates a complete and production-ready application.

The credit-based SaaS model demonstrates the commercial viability of the platform, making it accessible to a wide range of students while providing a sustainable business model. The PDF download feature, notes history panel and quick revision mode further enhance the utility and value of the platform for students preparing for examinations.

The implementation of the **AI Exam Notes Generator** contributes to improved student productivity, smarter exam preparation and reduced academic stress. It showcases the power of artificial intelligence in the education sector and highlights the importance of combining technology with learning to create meaningful and impactful solutions.

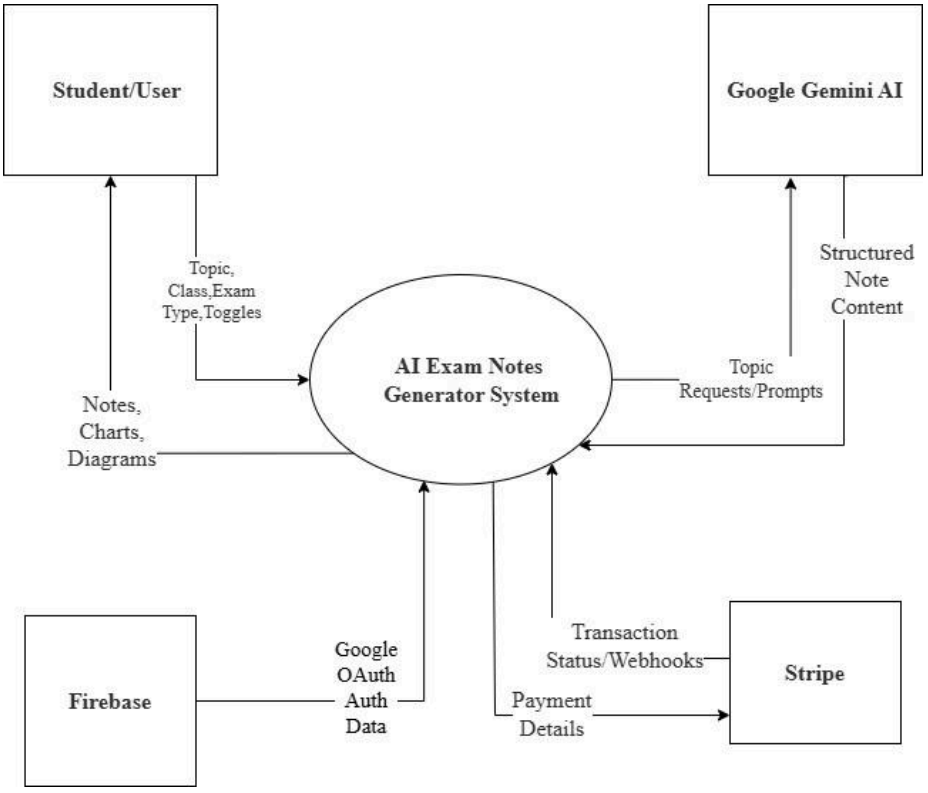
In the future, the system can be enhanced by adding features such as mobile application support, AI-generated mock tests, performance analytics, collaborative note sharing, regional language support and integration with learning management systems. These improvements will further increase the scalability, usability and reach of the platform. Overall, the project successfully fulfills its objectives and demonstrates the effective application of MERN stack, AI and SaaS design principles.

DATA FLOW DIAGRAM (DFD) – LEVEL 0

The **Data Flow Diagram (DFD) Level 0,** also known as the **Context Diagram**, represents the overall structure of the **AI Exam Notes Generator** system as a single process. It shows how the system interacts with all external entities and illustrates the high-level flow of data between the system and the outside world without showing internal processes in detail.

The external entities that interact with the **AI Exam Notes Generator** system are: Student/User who provides the topic, class level, exam type and feature toggles and receives generated notes, charts, diagrams and PDF output; Google Gemini AI which receives structured topic prompts from the system and returns AI-generated note content; Firebase which handles Google OAuth authentication data; and Stripe which processes payment details and sends back transaction status and webhook confirmations.

DATA FLOW DIAGRAM (DFD) – LEVEL 0 AI EXAM NOTES GENERATOR SYSTEM



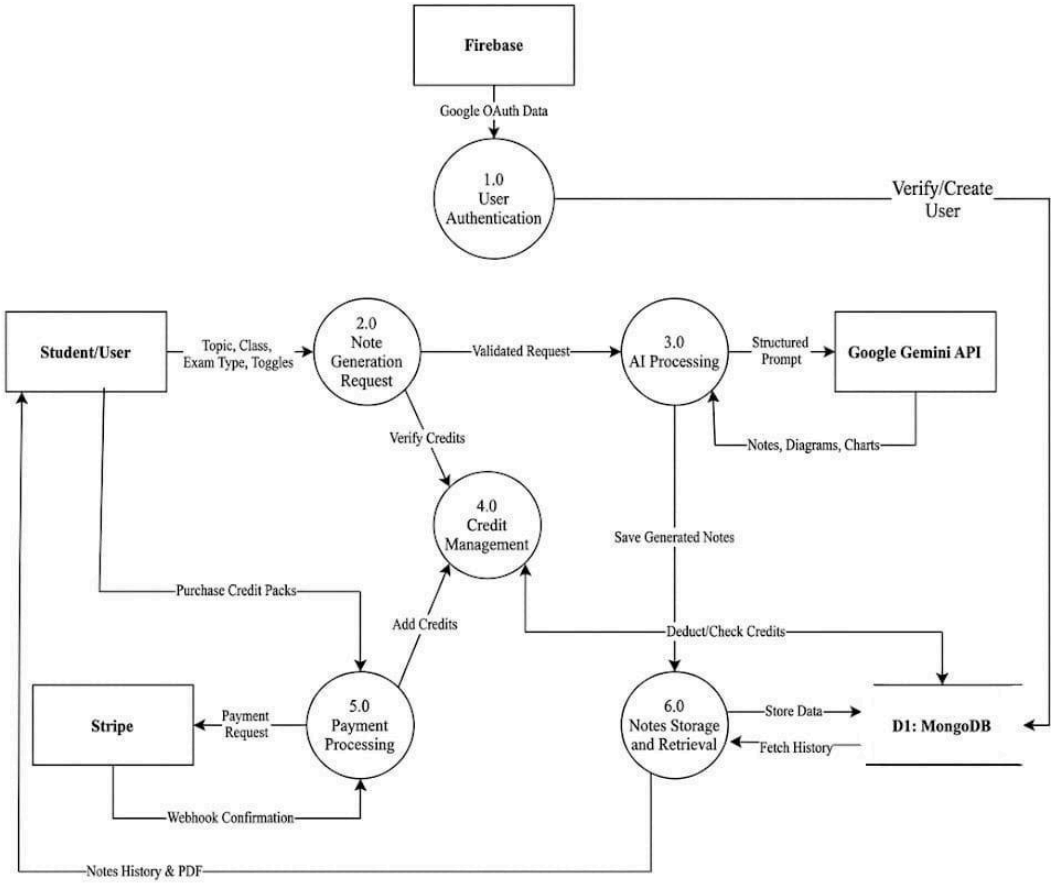
The Context Diagram above represents the entire **AI Exam Notes Generator** system as a single central process. It clearly illustrates the interaction between the system and all four external entities, showing the input and output data flows that define the system boundary.

DATA FLOW DIAGRAM (DFD) – LEVEL 1

The **Data Flow Diagram (DFD) Level 1** provides a detailed breakdown of the **AI Exam Notes Generator** into six key sub-processes. It explains how data moves between different components of the system and illustrates the internal working of each major function step by step.

The **Level 1 DFD** divides the main system into: 1.0 User Authentication which handles Google OAuth login via Firebase; 2.0 Note Generation Request which receives topic, class, exam type and feature toggles from the user; 3.0 AI Processing which sends structured prompts to the Gemini API and returns notes, diagrams and charts; 4.0 Credit Management which verifies, deducts and adds credits; 5.0 Payment Processing which handles Stripe payments and webhook confirmations; and 6.0 Notes Storage and Retrieval which stores and fetches notes and history from MongoDB.

DATA FLOW DIAGRAM (DFD) – LEVEL 1 AI EXAM NOTES GENERATOR SYSTEM

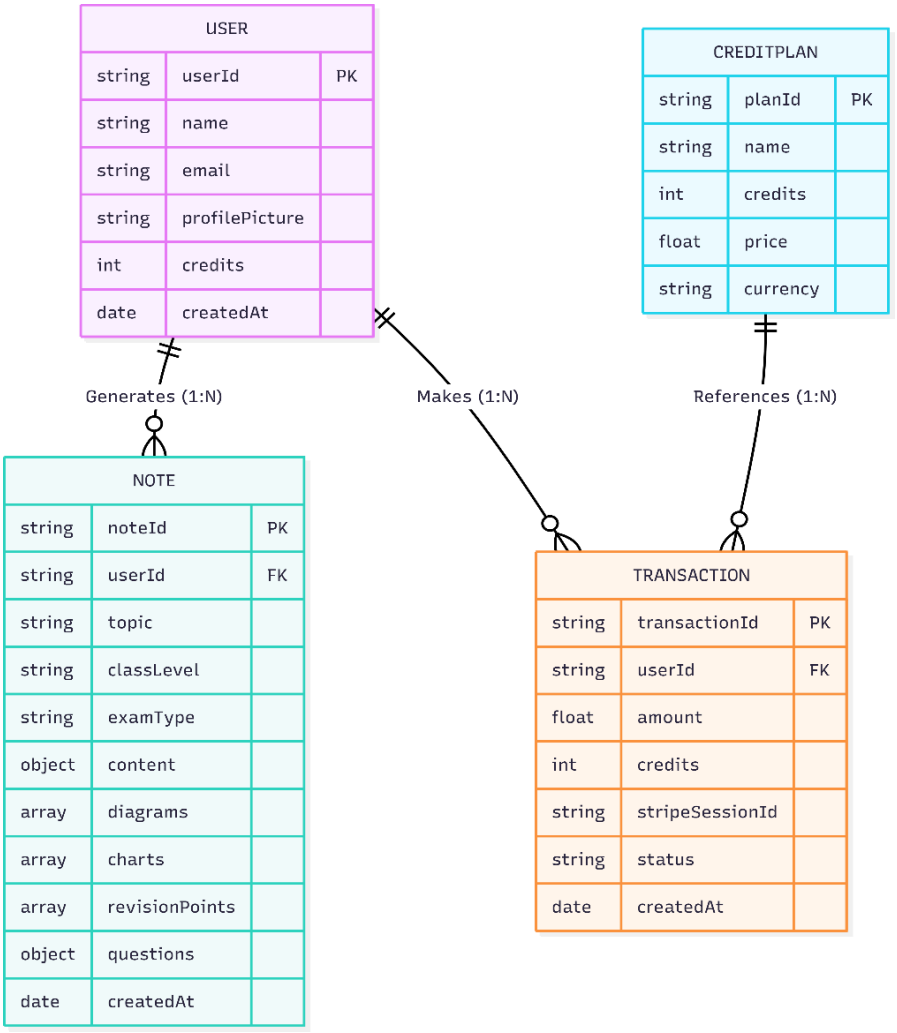


The **Level 1** Data Flow Diagram above provides a clear and detailed representation of how the AI Exam Notes Generator operates internally. It shows the interaction between all six processes, the MongoDB data store and the external entities including the Student/User, Google Gemini AI, Firebase and Stripe.

ER DIAGRAM (ENTITY RELATIONSHIP DIAGRAM)

In the **AI Exam Notes Generator**, the main entities include **User, Note, Transaction and CreditPlan.** These entities interact with each other to perform various operations such as generating notes, managing credits and processing payments. The ER diagram ensures data consistency and efficient data retrieval within the MongoDB database.

ER DIAGRAM (ENTITY RELATIONSHIP DIAGRAM) AI EXAM NOTES GENERATOR SYSTEM



The **Entity Relationship Diagram** above provides a clear understanding of how different entities in the **AI Exam Notes Generator** system are connected. It helps in designing a well- structured MongoDB database schema and ensures smooth system operation with accurate and reliable data management.

ENTITIES, ATTRIBUTES and RELATIONSHIPS

1. Entities

An Entity represents a real-world object or concept about which data is stored in the system. In the AI Exam Notes Generator, the main entities are:

- User
- Note
- Transaction
- CreditPlan

2. Attributes of Entities

Attributes are the properties or characteristics of an entity. The following tables define the attributes for each entity:

Entity	Attributes	Data Type	Primary Key	Foreign Key
User	_id (userId)	ObjectId	Yes	
	name	String	No	
	email	String	No (Unique)	
	profilePicture	String	No	
	credits	Number	No	
	createdAt	Date	No	
Note	_id (noteId)	ObjectId	Yes	
	userId	ObjectId ref	No	Yes → User
	topic	String	No	
	classLevel	String	No	
	examType	String	No	
	content	Object	No	
	diagrams	Array	No	
	charts	Array	No	
	revisionPoints	Array	No	
	questions	Object	No	
	createdAt	Date	No	
Transaction	_id (transactionId)	ObjectId	Yes	
	userId	ObjectId ref	No	Yes → User
	amount	Number	No	
	credits	Number	No	

	stripeSessionId	String	No
	status	String	No
	createdAt	Date	No
CreditPlan	_id (planId)	ObjectId	Yes
	name	String	No
	credits	Number	No
	price	Number	No
	currency	String	No

3. Relationships

- **User Note:** One-to-Many (one user can generate many notes)
- **User Transaction:** One-to-Many (one user can have many payment transactions)
- **Transaction CreditPlan:** Many-to-One (many transactions can reference one credit plan)

DATABASE STRUCTURE

The database structure defines the collections used in the MongoDB database along with their fields, data types, sizes and key constraints. It ensures proper data organization, storage and retrieval for efficient system performance.

1. USERS COLLECTION

Field Name	Data Type	Field Size	Primary Key	Foreign Key	Description
_id (userId)	ObjectId	Auto	Yes	No	Unique user ID
name	String	255 chars	No	No	Full name
email	String	255 chars	No (Unique)	No	Email address
profilePicture	String	512 chars	No	No	Profile image URL
credits	Number	Default:100	No	No	Available credits
createdAt	Date	Auto	No	No	Registration date

2. NOTES COLLECTION

Field Name	Data Type	Field Size	Primary Key	Foreign Key	Description
_id (noteId)	ObjectId	Auto	Yes	No	Unique note ID
userId	ObjectId ref	Auto	No	Yes → users	Reference to user
topic	String	255 chars	No	No	Note topic
classLevel	String	100 chars	No	No	Class / year
examType	String	100 chars	No	No	Exam board
content	Object	Variable	No	No	AI note content
diagrams	Array	Variable	No	No	Diagram data
charts	Array	Variable	No	No	Chart data
revisionPoints	Array	Variable	No	No	Revision points
questions	Object	Variable	No	No	Q&A sets
createdAt	Date	Auto	No	No	Created date

3. TRANSACTIONS COLLECTION

Field Name	Data Type	Field Size	Primary Key	Foreign Key	Description
_id (transId)	ObjectId	Auto	Yes	No	Unique trans. ID
userId	ObjectId ref	Auto	No	Yes → users	Reference to user
amount	Number	Decimal	No	No	Payment amount
credits	Number	Integer	No	No	Credits purchased
stripeSessionId	String	255 chars	No	No	Stripe session ID
status	String	20 chars	No	No	Paid / Pending
createdAt	Date	Auto	No	No	Transaction date

4. CREDITPLANS COLLECTION

Field Name	Data Type	Field Size	Primary Key	Foreign Key	Description
_id (planId)	ObjectId	Auto	Yes	No	Unique plan ID
name	String	100 chars	No	No	Plan name
credits	Number	Integer	No	No	Credits in plan
price	Number	Decimal	No	No	Plan price
currency	String	10 chars	No	No	Currency code

There are a total 4 collections in the **AI Exam Notes Generator MongoDB database: users, notes, transactions and credit plans**. Primary Keys ensure unique identification of documents, while Foreign Keys (referencing Object IDs) maintain relationships between collections. This structured database design supports efficient note generation, secure payment handling and proper management of user data and credit balances.

REFERENCES / BIBLIOGRAPHY

The successful completion of the **AI Exam Notes Generator** project was made possible with the help of various books, websites, online documentation and video resources. These references provided valuable knowledge about MERN stack development, AI API integration, payment gateway implementation and SaaS product design.

Books

- Flanagan, David. JavaScript: The Definitive Guide. O'Reilly Media.
- Duckett, Jon. HTML and CSS: Design and Build Websites. John Wiley & Sons.
- Banker, Kyle. MongoDB in Action. Manning Publications.
- Pressman, Roger S. Software Engineering: A Practitioner's Approach. McGraw- Hill Education.
- Sommerville, Ian. Software Engineering. Pearson Education.

Websites and Documentation

- www.reactjs.org Official React.js documentation
- www.nodejs.org Official Node.js documentation
- www.expressjs.com Express.js framework documentation
- www.mongodb.com/docs MongoDB official documentation
- www.ai.google.dev Google Gemini AI API documentation
- www.stripe.com/docs Stripe payment gateway documentation
- www.firebase.google.com/docs Firebase documentation
- www.w3schools.com Web development reference
- www.geeksforgeeks.org Programming and CS concepts
- www.mdn.web.com Mozilla Developer Network web docs

INTERNSHIP DETAILS

The following internship details are provided for both students as per the mandatory requirement for BCA 6th Semester Project Synopsis submission.

Student 1: Tanushree Dutta (Roll No: 2303265)

Company Name	Ardent Computech Pvt. Ltd.
Company Address	Module 103, SDF Building, Sector V, Salt Lake, Kolkata – 700091
Email	internships@ardentsoftware.in
Contact Number	033-40073507
Duration of Internship	August 22, 2025 to November 01, 2025 (2.3 months)
Internship Start Date	August 22, 2025
Internship End Date	November 01, 2025
Supervisor Name	Indranil De Sarkar
Supervisor Designation	Director – HR, Ardent Software Services
Nature of Work / Training	Data Science, Artificial Intelligence and Machine Learning Using Python; Project: Used Bike Price Prediction
Certificate ID	ARDENT/172128 (Industrial Internship) | ARDENT/172130 (Industrial Training)

Student 2: Payal Mahakur (Roll No: 2303163)

Company Name	TechCoder Software (OPC) Pvt. Ltd.
Company Address	H/CC-14 Ground Floor, Sidhgora Main Road, Champiya Colony, Jamshedpur, Jharkhand – 831009
Email	hr@techcoderservices.in
Contact Number	7667988362, 7667094447
Duration of Internship	15th May 2025 to 15th July 2025 (2 months)
Internship Start Date	15th May 2025
Internship End Date	15th July 2025
Supervisor Name	Shubham Kumar Singh
Supervisor Designation	Director, TechCoder Software (OPC) Pvt. Ltd.
Nature of Work / Training	Data Analytics; Project: Zomato Dashboard
Reference Number	TSS/HRD/INTR/118/2025

INTERNSHIP CERTIFICATE

The following internship certificates issued by the respective companies are attached as mandatory evidence of completion of the industrial training requirement for BCA 6th Semester:

Tanushree Dutta Certificate Attached:

Industrial Internship Certificate - Certificate ID: ARDENT/172128, issued November 01, 2025, by Ardent Computech Pvt. Ltd. (signed and stamped by Director - Technology Services and Director - Operations), certifying completion of project titled "Used Bike Price Prediction" under the technical team of Ardent Computech Pvt. Ltd.

Industrial Internship Certificate

ARDENT COMPUTECH PVT. LTD.

Ardent Verified

This certificate is awarded to

TANUSHREE DUTTA

of

NETAJI SUBHAS UNIVERSITY

for successfully completing the **Industrial Internship** on

DATA SCIENCE, ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING USING PYTHON

from

AUGUST 22, 2025 to NOVEMBER 01, 2025

and implementing the project titled

USED BIKE PRICE PREDICTION

Certificate ID: ARDENT/172128
Issue Date: NOVEMBER 01, 2025

Director Technology Services

Director Operations

Payal Mahakur Certificate Attached:

Internship Completion Certificate - Reference: TSS/HRD/INTR/118/2025, dated July 15, 2025, issued by TechCoder Software (OPC) Pvt. Ltd., certifying completion of Data Analytics internship and project titled "Zomato Dashboard" (signed and stamped by Shubham Kumar Singh, Director).

TECHCODER SOFTWARE (OPC) PVT LTD

AFFORDABLE INNOVATION, EXCEPTIONAL RESULTS.

CERTIFICATE OF INTERNSHIP

This **Data Analytic** Internship program certificate is proudly awarded to

Miss. Payal Mahakur

For her outstanding completion of the compulsory Internship program at TechCoder Software (OPC) Pvt. Ltd

15th May 2025 to **15th July 2025**

15/07/2025

DATE

DIRECTOR

H/CC-14, Main Road Champiya Colony Sidhgora, Jamshedpur, Jharkhand

www.techcoderservice.in, hr@techcoderservice.in

(Photocopies of all certificates are attached on the following pages as per the submission guidelines)